Course Number: CSE2023	**Course Name:** Object Oriented Programming using Java	
Credits: 3 (2 L – 0 T – 2 P)	**Contact Hours per week:** 2 hours theory & 2 hours lab per week	
Batch: 2024-2028, 3rd Sem **Academic Year:** 2024-2025	**Semester Duration**: 28th Jul 2025 ***to*** 28th Nov 2025	
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Aim of the course:

The aim of this course is to help the students understand the concepts of Object-Oriented Programming (OOP). At the end of the course, the students will be able to develop efficient programs using object-oriented features in Java and able to solve complex real world problems using Java.

Course Overview and Context:

This course is designed to provide a comprehensive foundation in Java programming, covering essential concepts and techniques for building robust and efficient applications. This course is tailored for beginners and intermediate programmers who wish to gain a solid understanding of Java, which is one of the most widely used programming languages in the industry.

Through this course, learners will explore Java's fundamental principles, including object-oriented programming (OOP), data types, control structures, exception handling, multithreading, byte/character streams and collection framework libraries along with core java libraries. By the end of the course, participants will be equipped to develop, debug, and maintain Java applications and will have a strong foundation to progress to advanced Java topics and frameworks.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO1	Understand and apply object-oriented programming concepts using Java.	Skill Development & Employability	Written Tests and Quizzes, Conceptual and Written Assignments and Viva
CO2	Develop Java programs using core language features, including control structures, classes, interfaces, and exception handling.	Skill Development, Employability & Entrepreneurship	Lab Assignments, Practical Exams, Instructor Lead Code Reviews Sessions, Hands-on Lab Tasks, Mid-Term / End-Term Practical Exams
CO3	Design and implement basic programs/applications using file handling, multithreading, and collections framework in Java.	Skill Development, Employability & Entrepreneurship	Mini Projects, Demos, Hackathons, Time Bound Lab Problems

Prerequisite: if any

1. Fundamentals of Computer

Topics of the course:

Topics	Number of Sessions	Course Outcomes
Characteristics of Java, the Java environment – JDK, JRE, JVM and JIT Compiler, Object Oriented Programming (OOP) principles – OOP concepts in Java, Lifecycle of a Java program – Java Byte Code, First Java program in Java.	4 (3 Theory+1 Lab)	CO1
Fundamental Programming Structures in Java – Identifiers, Literals, Comments, Data Types, Variables – Scope and lifetime of a variable, and type conversion, Operators, Expressions, Control Flow – conditional statements, switch, loops, and jump statements.	7 (5 Theory+2 Lab)	CO2
Defining classes in Java, Creating an Object, methods, constructors, Overloading – method overloading and constructor overloading, this keyword, static members, nested classes, Packages, Arrays, Strings, Scanner class, Vectors, Wrapper classes.	7 (4 Theory+3 Lab)	CO1, CO2
Inheritance – Super classes and sub classes, access modifiers, super keyword, overriding – method overriding, constructors in sub classes, dynamic method dispatch, abstract classes and methods, final keyword, Interface – defining and implementing an interface, extending the interfaces.	6 (4 Theory+2 Lab)	CO1, CO2
Exceptions – errors vs exceptions, types of exceptions – built-in and user-defined exceptions, throwing and catching exceptions – try/catch block, finally block, throw and throws keywords, creating and throwing own exceptions.	6 (4 Theory+2 Lab)	CO3
Multithreading- thread-based vs process-based multitasking, creating threads using Thread class and Runnable interface, fields and methods of Thread class, thread priorities, thread lifecycle methods, synchronizing threads, inter thread communication.	6 (4 Theory+2 Lab)	CO3
I/O Basics – Streams, Byte streams and Character streams, Reading and Writing Console, Reading and Writing Files, The Serializable Interface, Creating files and directories.	6 (4 Theory+2 Lab)	CO3
Collection Framework – List - ArrayList, Linked List and Vector, Set – HashSet and TreeSet, Map – HashTable and HashMap, Accessing a Collection via an iterator, comparator, comparable.	6 (4 Theory+2 Lab)	CO3
Total Contact Sessions:	**(32 Theory & 16 Lab) = 48**	

CO/PO Mapping:

Course (Level) Mapping		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
Mapping of Course Outcomes	CO1	2	3	1		1						1					
	CO2	3	2	1	1	2						2	1	1	1		
	CO3	3	2	1	3	3						2	1	3	2		

Learning Resources:

Textbook:

1. Herbert Schildt. Java: The Complete Reference. 12th Edition, McGraw Hill Education; 2022.
2. Horstmann Cay S., Core Java Volume II – Advanced Features, Pearson, 2019

Reference Books:

1. Khalid A. Mughal and Rolf W Rasmussen, Programmer's Guide to Java SE 8 Oracle Certified Associate (OCA), 1st Ed, Pearson, 2017
2. E. Balagurusamy, Programming with Java – A Primer, 6th Ed, TMH India, 2019

Experiential Learning Component:

The Experiential Learning Component of the "Object Oriented Programming using Java" course (CSE2023) is designed to provide students with hands-on experiences that reinforce theoretical knowledge through practical application. Throughout the course, students will engage in lab exercise-based learning where they will be assigned practical questions that evolve from basic java programs to the advanced level applications. These exercises, which span the entirety of the course, are structured into phase-wise evaluations to ensure consistent application and progression of skills.

Experiential Learning Percentage: 90%

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the two lab evaluations and an end semester exam along with one quiz. Relative grading methods defined in the academic regulations of the university will be followed to grade the students. Students must secure a minimum 40% mark after completing all the assessments in the following table to become eligible for grading.

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics	Course Outcome
Quiz	20%	In the week starting from 22nd September	Topics to be covered will be announced in the class. 20 MCQ questions, each of 1 mark. Mode of quiz will be offline.	CO1, CO2
Lab Evaluation 1	30%	Will be held throughout the week starting from 20st October	**Focus:** Programming questions based on Java fundamentals learn till the date of lab evaluation. – Three java programs will be given in the exam, where each question will carry 10 marks. – All questions would be compulsory. – Writeup of each question will fetch 5 marks. – Computer execution of each question will also fetch the 5 marks.	CO1, CO2
Lab Evaluation 2	10%	Throughout the Semester: Report will be submitted in the last week of November	An indexed and date-wise lab report consisting of the solutions of lab exercises given in the lab sessions.	CO1, CO2, CO3
End Semester Exam	40%	As per academic calendar/Date-sheet	Close book written exam based on entire syllabus mentioned the course file.	CO1, CO2, CO3

– All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code. – **Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity.**	–

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.